

PAGE 1 · THE COMPANY PROBLEM

Measurement must become qualified action.

4MP is building a new technological foundation for precision manufacturing: CNC machines that self-measure, self-correct, and improve from production evidence.

The VP of AI must build the intelligence layer and the founding technical organization that can turn digital intent, physical variation, and metrology into trustworthy corrective action.

Why this matters now

- Setups, measurements, and manual fixes keep expensive machines waiting.
- Digital models assume ideal machines, tools, fixtures, and conditions.
- Agile, distributed manufacturing needs first-pass quality without rare expertise.
- Every correction can become a reusable asset only when context and evidence remain attached.

Four decisions must be explicit

- What may act now?
- What explains the measured difference?
- What evidence proves the result?
- What may generalize beyond this machine, tool, part, and environment?

Operating thesis

Make every correction explainable, bounded, validated, and reusable.

First proof

Choose one valuable correction class, close it end to end in shadow mode, stress its limits, and earn a bounded release posture.

THE CORRECTION ARCHITECTURE

INTENT

What should happen

STATE

What is happening

CAUSE

Why it differs

BOUND

What may change

PROVE

Did it improve

MEMORY

What may be reused

One correction. Three systems that mature together.

Intelligence

- Recover manufacturing intent from G-code, CAM, geometry, tolerances, and operator context.
- Reconstruct machine, tool, part, fixture, material, environment, and time.
- Separate measured residual from ranked cause hypotheses.
- Propose correction locus, alternatives, confidence, and expected consequences.

Evidence

- Replay and simulation before machine experimentation.
- Shadow comparison against expert decisions.
- Controlled test with limits, fallback, and rollback.
- Post-action metrology, repeatability, recurrence, and secondary effects.

Organization

- Technical ownership across AI/ML, geometry, physics, data, machine integration, metrology, and platform software.
- Architecture review that records alternatives and test implications.
- Correction review that connects intent, state, cause, authority, evidence, and disposition.
- Hiring and roadmap decisions based on correction-class leverage.

Data foundation

- Canonical correction object and coordinate-frame lineage.
- Model, physics, rule, data, and software versions.
- Proposed versus executed action and named decision owner.
- Reuse scope, expiry, and incident history.

Authority should expand by qualified correction class—not by broad model confidence.

EVIDENCE TRANSFER

Vape-Jet

Robotics, machine vision, approximately 1,000 fielded robots, and cross-functional machine/factory/customer feedback loops.

Compunetics

Advanced manufacturing, high-reliability programs, quality systems, root cause, process controls, and cross-domain technical leadership.

DudeWorth

Agentic workflows, structured context, human review, evaluation, escalation, governance, and adoption.

ZeusVu + Amazon

Regulated autonomous operations, safety boundaries, launch mechanisms, standard work, and scale.

Questions that expose the real correction system.

- 01 Which correction class currently offers the best combination of customer value, measurable proof, and constrained consequence?
- 02 Where does 4MP currently draw the boundary among learned models, physics-based models, deterministic rules, simulation, and human judgment?
- 03 What is the hardest coordinate-frame, registration, or lineage problem the team has encountered across CAD/CAM, CNC, sensors, and metrology?
- 04 What evidence must exist before a recommendation may become a bounded machine action?
- 05 Which failure modes would be most damaging to technical credibility with early customers or machine-tool partners?
- 06 What parts of the current architecture are durable platform contracts, and what remains intentionally experimental?
- 07 Which first hires would unlock the next two correction classes rather than duplicate the founders' strongest capabilities?
- 08 How should customer evidence, technical proof, and product roadmap decisions meet in the same operating cadence?

PROPOSED 90-MINUTE TECHNICAL WORKING SESSION

Frame

One representative correction class, expected value, and consequence boundary.

Trace

Intent, state, residual, cause, action, authority, proof, and reuse scope.

Stress

Missing data, program ambiguity, coordinate error, model disagreement, rollback, and incident response.

Decide

Shadow, bounded release, hold, next experiment, and architecture implications.

The goal is not to present a finished answer. It is to demonstrate how I would work with 4MP's technical leaders on an unsolved physical-AI problem.